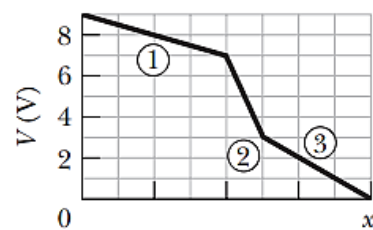


Ch26: 4, 5, 27, 30, 46

Basic score 4` + problem 4 3`+ problem 27 3`

4. The absolute values of the slopes (for the straight-line segments shown in the graph) are equal to the respective electric field magnitudes. Thus, we have:

$$\begin{aligned}\rho_1 J_1 &= \frac{J_1}{\sigma_1} = E_1 = -\frac{\Delta V_1}{\Delta x_1} = \frac{2\text{V}}{4.0\text{mm}} = 500 \text{ V/m} \\ \rho_2 J_2 &= \frac{J_2}{\sigma_2} = E_2 = -\frac{\Delta V_2}{\Delta x_2} = \frac{4\text{V}}{1.0\text{mm}} = 4000 \text{ V/m} \\ \rho_3 J_3 &= \frac{J_3}{\sigma_3} = E_3 = -\frac{\Delta V_3}{\Delta x_3} = \frac{3\text{V}}{3.0\text{mm}} = 1000 \text{ V/m}\end{aligned}$$



0.5 point each, 1.5 point in total

或者如果答案写成如下形式，则按：

$$J = \sigma E \quad 0.5 \text{ point}$$

$$E = -\Delta V / \Delta x \quad \text{这里只求 } E \text{ 的大小，没有负号不用扣分} \quad 0.5 \text{ point}$$

$$\frac{E_3}{E_1} = 2, \quad \frac{E_3}{E_2} = 1/4 \quad (\text{可后面公式再给出}) \quad 0.5 \text{ point}$$

We note that the current densities are the same since the values of i and A are the same (see the problem statement) in the three sections, so $J_1 = J_2 = J_3$. 0.5 point

(a) Thus we see that

$$\sigma_1 E_1 = \sigma_3 E_3 \Rightarrow \sigma_1 = \frac{E_3}{E_1} \sigma_3 = 2\sigma_3 = 8.00 \times 10^7 (\Omega \cdot \text{m})^{-1} \quad 0.5 \text{ point (结果分)}$$

(b) Similarly,

$$\sigma_2 E_2 = \sigma_3 E_3 \Rightarrow \sigma_2 = \frac{E_3}{E_2} \sigma_3 = \frac{\sigma_3}{4} = 1.00 \times 10^7 (\Omega \cdot \text{m})^{-1} \quad 0.5 \text{ point (结果分)}$$

5. (a) The current i is shown in Fig. 26-29 entering the truncated cone at the left end and leaving at the right. This is our choice of positive x direction. We make the assumption that the current density J at each value of x may be found by taking the ratio i/A where $A = \pi r^2$ is the cone's cross-section area at that particular value of x .

The direction of $\vec{J}$ is identical to that shown in the figure for i (our $+x$ direction).

Using Eq. 26-11, we then find an expression for the electric field at each value of x , and next find the potential difference V by integrating the field along the x axis, in accordance with the ideas of Chapter 25. Finally, the resistance of the cone is given by $R = V/i$. Thus,

$$J = \frac{i}{\pi r^2} = \frac{E}{\rho}$$

where we must deduce how r depends on x in order to proceed. We note that the radius increases linearly with x , so (with c_1 and c_2 to be determined later) we may write

$$r = c_1 + c_2 x.$$

Choosing the origin at the left end of the truncated cone, the coefficient c_1 is chosen so that $r = a$ (when $x = 0$); therefore, $c_1 = a$. Also, the coefficient c_2 must be chosen so that (at the right end of the truncated cone) we have $r = b$ (when $x = L$); therefore, $c_2 = (b - a)/L$. Our expression, then, becomes

$$r = a + \left(\frac{b - a}{L} \right) x.$$

Substituting this into our previous statement and solving for the field, we find

$$E = \frac{i\rho}{\pi} \left(a + \frac{b - a}{L} x \right)^{-2}.$$

Consequently, the potential difference between the faces of the cone is

$$\begin{aligned} V &= - \int_0^L E dx = - \frac{i\rho}{\pi} \int_0^L \left(a + \frac{b - a}{L} x \right)^{-2} dx = \frac{i\rho}{\pi} \frac{L}{b - a} \left(a + \frac{b - a}{L} x \right)^{-1} \Bigg|_0^L \\ &= \frac{i\rho}{\pi} \frac{L}{b - a} \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{i\rho}{\pi} \frac{L}{b - a} \frac{b - a}{ab} = \frac{i\rho L}{\pi ab}. \end{aligned}$$

The resistance is therefore

$$R = \frac{V}{i} = \frac{\rho L}{\pi ab} = \frac{(731 \Omega \cdot \text{m})(3.50 \times 10^{-2} \text{ m})}{\pi(1.70 \times 10^{-3} \text{ m})(2.30 \times 10^{-3} \text{ m})} = 2.08 \times 10^6 \Omega$$

Alternatively, we can apply the follow integration to get R directly:

$$R = \int_{\text{cone}} dR = \int_{\text{cone}} \frac{dV}{i} = \int_{\text{cone}} \frac{E dx}{JA} = \frac{\rho}{\pi} \int_0^L \left(a + \frac{b - a}{L} x \right)^{-2} dx = \frac{\rho}{\pi} \frac{L}{b - a} \left(a + \frac{b - a}{L} x \right)^{-1} \Bigg|_0^L = \frac{\rho L}{\pi ab}.$$

27. (a) The current in the block is $i = V/R = 35.8 \text{ V}/935 \Omega = 3.83 \times 10^{-2} \text{ A}$.

0.5 mark 公式 0.3 mark, 结果 0.2 mark

(b) The magnitude of current density is

$$J = i/A = (3.83 \times 10^{-2} \text{ A})/(2.70 \times 10^{-4} \text{ m}^2) = 142 \text{ A/m}^2.$$

0.5 mark 公式 0.3 mark, 结果 0.2 mark

(c) $v_d = J/ne = (142 \text{ A/m}^2)/[(5.33 \times 10^{22}/\text{m}^3)(1.60 \times 10^{-19} \text{ C})] = 1.67 \times 10^{-2} \text{ m/s}$.

1 mark 公式结果各 0.5 分

(d) $E = V/L = 35.8 \text{ V}/0.117 \text{ m} = 306 \text{ V/m}$. 1 mark 公式结果各 0.5 分

30. (a) Using Table 26-1 and Eq. 26-10 (or Eq. 26-11), we have

$$|\vec{E}| = \rho |\vec{J}| = (1.69 \times 10^{-8} \Omega \cdot \text{m}) \left(\frac{2.00 \text{ A}}{2.40 \times 10^{-6} \text{ m}^2} \right) = 1.41 \times 10^{-2} \text{ V/m}.$$

$$R = \rho L/A = 0.0282 \Omega, \quad V = iR, \quad E = V/L = 1.41 \times 10^{-2} \text{ V/m}$$

(b) Using $L = 4.0 \text{ m}$, the resistance is found from Eq. 26-16:

$$R = \rho L/A = 0.0282 \Omega.$$

The rate of thermal energy generation is found from Eq. 26-27:

$$P = i^2 R = (2.00 \text{ A})^2 (0.0282 \Omega) = 0.113 \text{ W}.$$

Assuming a steady rate, the amount of thermal energy generated in 30 minutes is found to be $(0.113 \text{ J/s})(30 \times 60 \text{ s}) = 2.03 \times 10^2 \text{ J}$.

46. (a) Circular area depends, of course, on r^2 , so the horizontal axis of the graph in Fig. 26-23(b) is effectively the same as the area (enclosed at variable radius values), except for a factor of π . The fact that the current increases linearly in the graph means that $i/A = J = \text{constant}$. Thus, the answer is “yes, the current density is uniform.”

(b) We find: $J = i/(\pi r^2) = (5.0 \text{ mA})/(\pi \times 8 \times 10^{-6} \text{ m}^2) = 199 \approx 2.0 \times 10^2 \text{ A/m}^2$.

$$(c) \quad J = \frac{i}{\pi r_s^2} = \frac{i'}{\pi r^2} \Rightarrow i' = \frac{r^2}{r_s^2} i = \frac{1}{2} i = 2.5 \text{ mA}$$